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PHY1001 Mechanics

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Bungee Jumping

A person of mass m exploits bungee jumping at at a cliff. The bungee cord has a length of L and a spring constant k. The person then jumps off the cliff with the cord tied, with an initial velocity $v_0 = 0$.
b) What is the maximum extension of the bungee cord?

Momentum

$K = \frac{1}{2}mv^2 = \frac{1}{2}\frac{(mv)^2}{m} = \frac{P^2}{2m}$

- $\overrightarrow{F} = m\frac{d\vec{v}}{dt} = \frac{d(m\vec{v})}{dt}$
- Define $\overrightarrow{P} = m\vec{v}$ as momentum.
  - Unit: $\mathrm{kg \cdot m/s}$
  - Momentum is a vector!
- Time rate of momentum is force, $\overrightarrow{F} = \frac{d\overrightarrow{P}}{dt}$
- Kinetic energy $K = \frac{P^2}{2m}$

Impulse

- Impulse $\vec{J} = \int \vec{F} dt$ (vector)
  - Unit: $\mathrm{kg \cdot m/s}$
- Change of momentum $\Delta \overrightarrow{P} = \vec{J}$

$\vec{J} = \int_{t_0}^{t_1} \vec{F}\, dt$ dimension analysis

unit $\rightarrow [\vec{J}] = [F \cdot t]$

$= kg \cdot m/s$

$\vec{F} = \frac{d\vec{p}}{dt}$

$\int_{t_0}^{t_1} \vec{F} dt = \int_{p_0}^{p_1} d\vec{p} = \Delta \vec{p}$

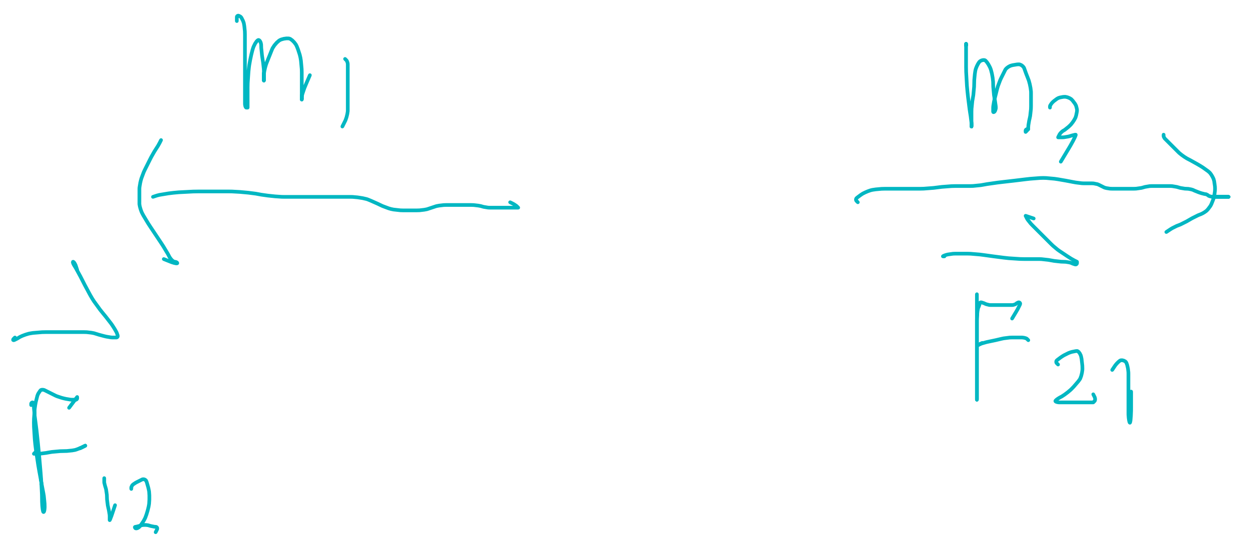
$\parallel$

$\vec{J}$

$\boxed{\vec{J} = \Delta \vec{p}}$

Conservation of Momentum

- If external forces are zero or negligible, then the momentum of the system is conserved!
  - $\sum \vec{P}_i = \text{const}\,.$
-



$$\vec{J}_{12} = \int \vec{F}_{12}\,dt = \Delta\vec{P}_1$$

$$\vec{J}_{21} = \int \vec{F}_{21}\,dt = \Delta\vec{P}_2$$

$$\Delta\vec{P} = \Delta\vec{P}_1 + \Delta\vec{P}_2$$

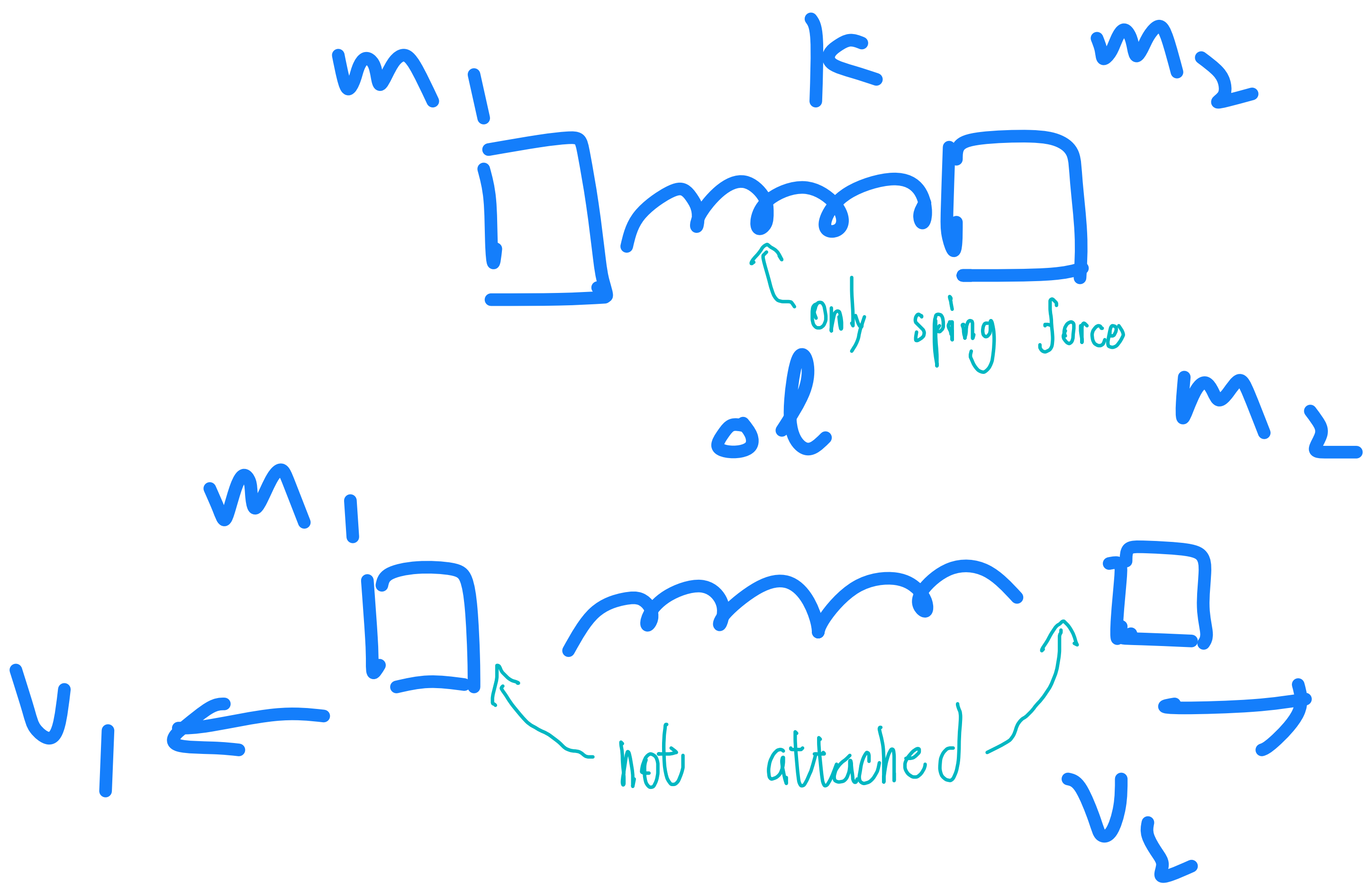
$$= \int_{t_0}^{t_1} \left(\vec{F}_{21} + \vec{F}_{12}\right) dt$$

$$\vec{F}_{21} = -\vec{F}_{12}$$

$$= 0 \Rightarrow \sum \vec{P}_i = \text{const.}$$

Exercise

A spring is compressed by length Δl between two objects. At one moment, the two objects are released. The two objects are not attached to the spring. The masses of the two objects are m_1 and m_2, respectively. The spring constant is k. Find the velocity of the two objects, v_1 and v_2, after release.



Momentum cons $\quad m_1 v_{1f} = m_2 v_{2f} \quad (1)$

ME cons $\quad \frac{1}{2} k \Delta l^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_2^2 \quad (2)$

$$\frac{1}{2} k \Delta l^2 = \frac{1}{2} m v_1^2 + \frac{(m_1 v_1)^2}{2 m_2}$$

$$k \Delta l^2 = m_1 \left(1 + \frac{m_1}{m_2}\right) v_1^2$$

$$v_1 = \sqrt{\frac{m_2 k}{m_1 (m_1 + m_2)}}\, \Delta l$$

$$v_2 = \frac{m_1}{m_2} v_1 = \sqrt{\frac{m_1 k}{m_2 (m_1 + m_2)}}\, \Delta l$$

Collision

- Collision is a process where two or more bodies exert forces on each other. Usually, the external forces are negligible compared to internal force, and thus momentum is conserved in a collision.
- Two types of collisions
  - Elastic collision
  - Inelastic collision
    - Complete inelastic collision

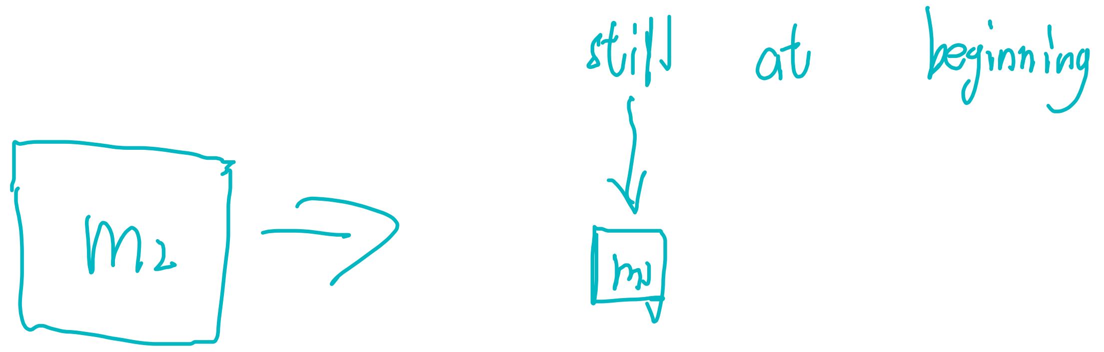
Elastic Collision

- Elastic collision: energy is conserved
- Consider two-body process:
  - Momentum is conserved: $m_1 v_{1,i} + m_2 v_{2,i} = m_1 v_{1,f} + m_2 v_{2,f}$
  - Energy is conserved: $\frac{1}{2}m_1 v_{1,i}^2 + \frac{1}{2}m_2 v_{2,i}^2 = \frac{1}{2}m_1 v_{1,f}^2 + \frac{1}{2}m_2 v_{2,f}^2$
  - Find $v_{1,f}$ and $v_{2,f}$ (HW4, q9)

m_1 $\xrightarrow{\quad v_{1i} \quad}$ m_2 $\xrightarrow{\quad v_{2i} \quad}$

Elastic Collision

- Consider an easier case: $v_{1,i} = 0$
  - Momentum is conserved: $m_2 v_{2,i} = m_1 v_{1,f} + m_2 v_{2,f}$
  - Energy is conserved: $\frac{1}{2} m_2 v_{2,i}^2 = \frac{1}{2} m_1 v_{1,f}^2 + \frac{1}{2} m_2 v_{2,f}^2$
  - Find $v_{1,f}$ and $v_{2,f}$
- What if $m_2 \gg m_1$? $V_{2f} = -V_{2i}$
- What if $m_2 \ll m_1$? $V_{2f} = V_{2i}$
- What if $m_1 = m_2$? $V_{2f} = 0$



$$\begin{cases} \text{P. cons} & m_2 V_{2i} = m_1 V_{1f} + m_2 V_{2f} \\ \text{KE. cons} & \frac{1}{2} m_2 V_{2i}^2 = \frac{1}{2} m_1 V_1^2 + \frac{1}{2} m_2 V_{2f}^2 \end{cases}$$

$$\begin{cases} m_2 (V_{2i} - V_{2f}) = m_1 V_{1f} & (3) \\ m_2 (V_{2i} - V_{2f})(V_{2i} + V_{2f}) = m_1 V_{1f}^2 & (4) \end{cases}$$

$\frac{(4)}{(3)}$ $V_{2i} + V_{2f} = V_{1f} \rightarrow (3)$

$$V_{1f} = \frac{2m_2}{m_1 + m_2} V_{2i} \qquad V_{2f} = \frac{m_2 - m_1}{m_1 + m_2} V_{2i}$$

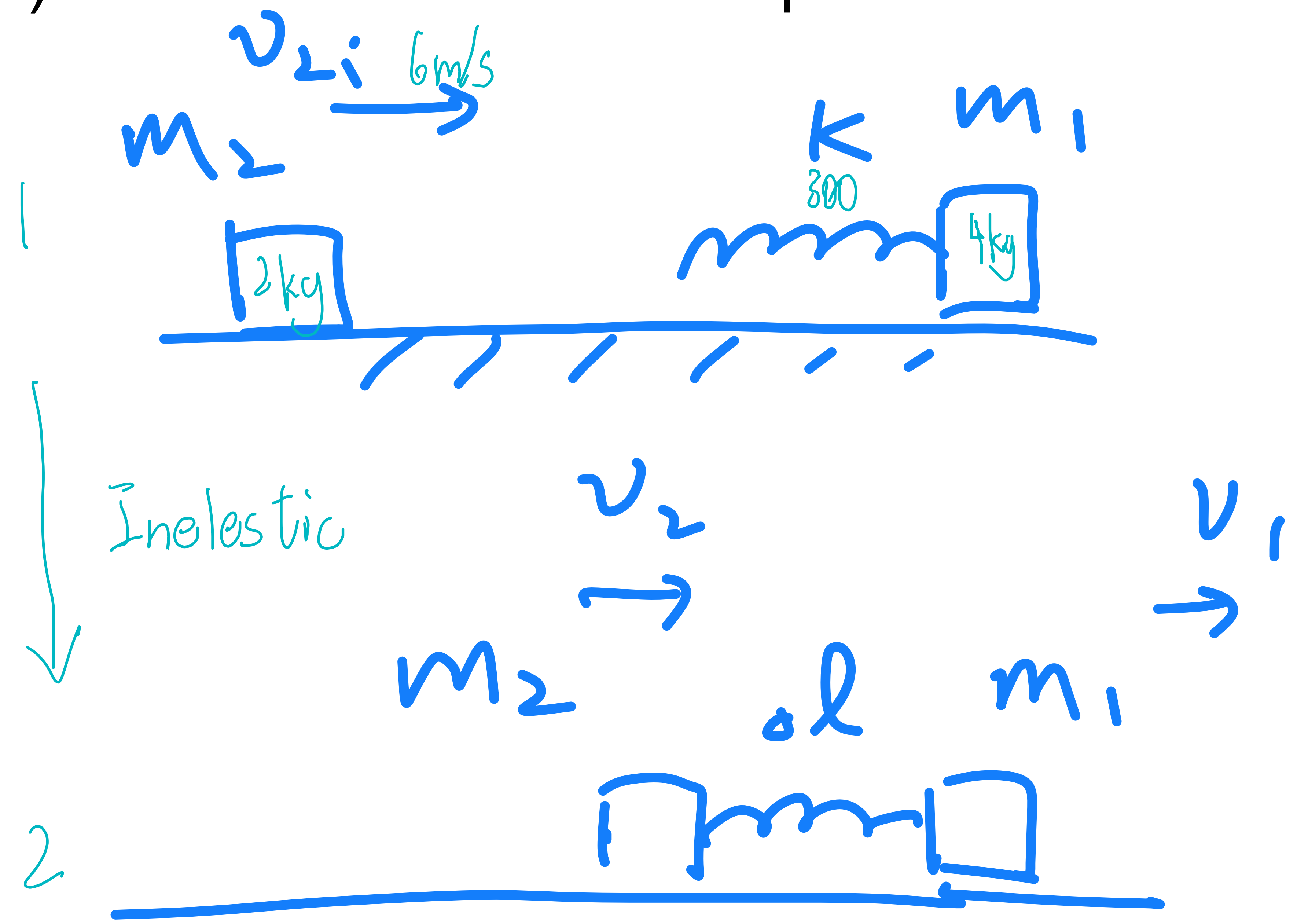
Inelastic Collision

- Inelastic collision: energy is NOT conserved
  - **Complete inelastic collision**: the velocity of each part is the same after collision
  - For a two-body process: $v_{1f} = v_{2f}$

Exercise

An object of mass $m_1 = 4\,\mathrm{kg}$is attached to a massless spring with a spring constant $k = 300\,\mathrm{N/m}$, and is at rest on a horizontal smooth surface. Another object $m_2 = 2\,\mathrm{kg}$ is moving right on the surface with a speed $v_{2i} = 6\,\mathrm{m/s}$and hits the spring, as shown in the figure below.

a) Find the maximum compression Δl



$$m_2 v_{2i} = m_1 v + m_2 v$$

$$= (m_1 + m_2) v$$

Exercise

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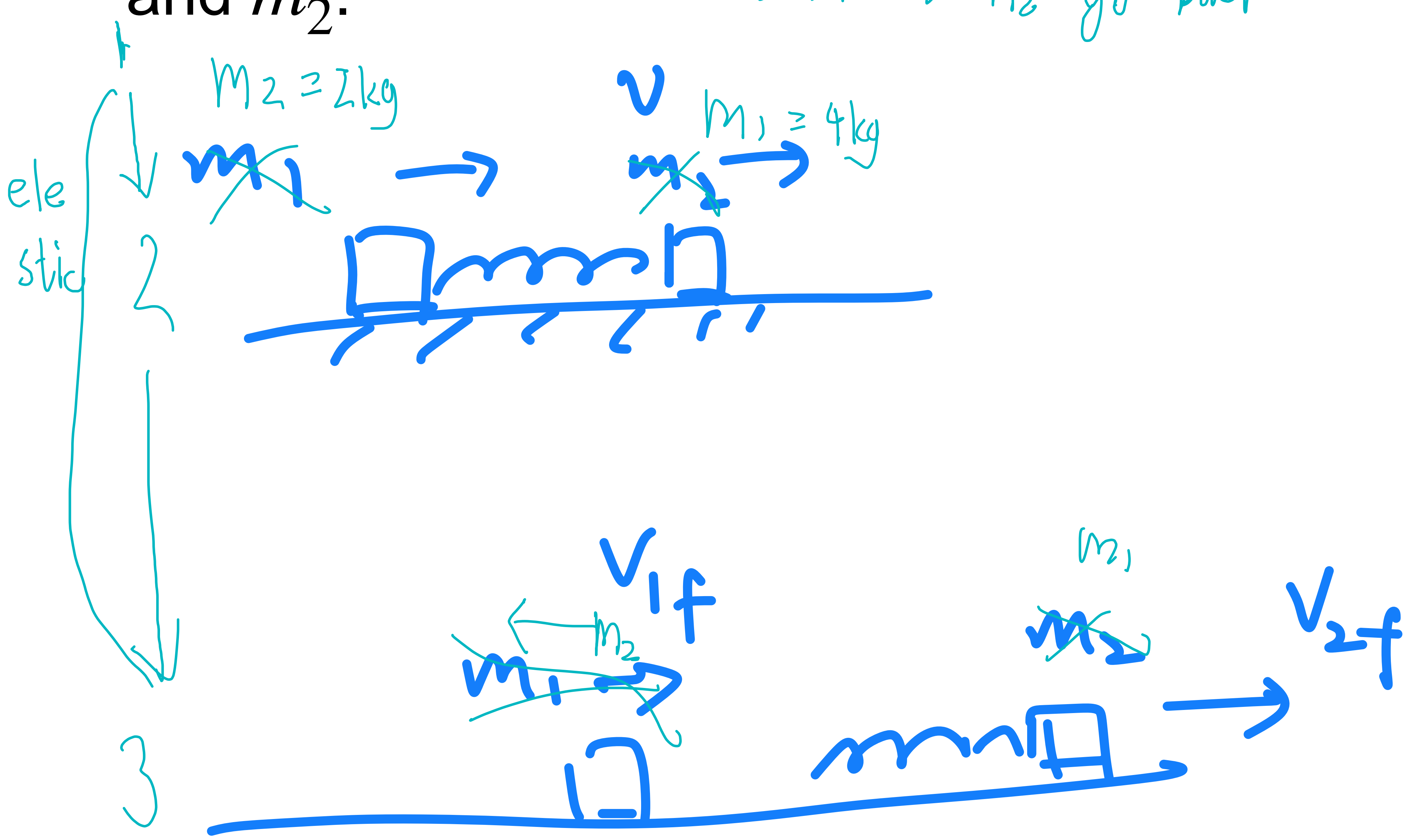
b) The spring then spring back, and m_2 and m_1 deattach eventually. Find the velocity v_{1f} and v_{2f} of m_1 and m_2.

$m_2 < m_1 \Rightarrow m_2$ go back



$$P: \; m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$KE: \; \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$$

$$v_{1f} = \frac{2m_2}{m_1 + m_2} v_2 = \frac{2 \cdot 2}{2+4} \cdot 6\,\mathrm{m/s} = 4\,\mathrm{m/s}$$

$$v_{2f} = \frac{m_2 - m_1}{m_1 + m_2} v_{2i} = \frac{-2}{2+4} \cdot 6\,\mathrm{m/s} = -2\,\mathrm{m/s}$$

$$KE_f = \frac{1}{2} \cdot 4 \cdot 4^2 + \frac{1}{2} \cdot 2 (-2)^2\,\mathrm{J} = 36\,\mathrm{J} \qquad KE_i = \frac{1}{2} \cdot 2 \cdot 6^2\,\mathrm{J} = 36\,\mathrm{J}$$